20261112 Quiz

for (MATH261) Discrete Mathematics

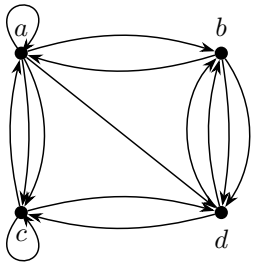
Jungwoo Kim
Department of Mathematics
Pohang University of Science and Technology

Problem 1.

(a) Draw the directed graph represented by the given adjacency matrix.

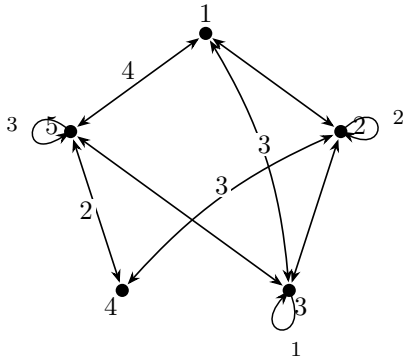
(b) Find the adjacency matrix of the given directed multigraph.

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 3 & 0 & 4 \\ 1 & 2 & 1 & 3 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ 0 & 3 & 0 & 0 & 2 \\ 4 & 0 & 1 & 2 & 3 \end{pmatrix}.$$



Solution.

(a) Since the matrix is symmetric, its off-diagonal entries give the same number of arcs in both directions. In the diagram, each bidirected edge represents a pair of opposite arcs; a label gives their multiplicity, and an unlabeled pair has multiplicity one.



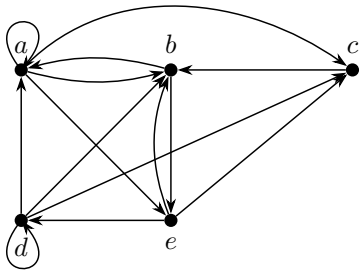
There are 2, 1, and 3 loops at vertices 2, 3, and 5, respectively.

(b) With rows and columns ordered as a, b, c, d, the adjacency matrix is

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \end{pmatrix}.$$

□

Problem 2. Which of the following are paths (starting at a vertex a and ending at a vertex b or a vertex c) in the following directed graph? What are the lengths of those that are paths? Find the circuits starting at a vertex a in the following directed graph.



Solution. The homework sheet does not provide the candidate vertex sequences referred to by "the following," so the requested path-by-path classification cannot be completed from the printed statement.

For the circuit question, recall that a circuit is a *closed trail*: its edges are distinct, although vertices may repeat. The directed edge set of the graph is

$$\{aa, ab, ac, ae, ba, be, cb, da, db, dc, dd, eb, ec, ed\},$$

where uv denotes the arc from u to v, and aa, dd denote loops. Every circuit starting at a is obtained by the following exhaustive procedure: start at a, successively follow any unused outgoing arc, and record the resulting trail whenever it returns to a. Continue exploring unused arcs after any such return, since a circuit may visit a more than twice.

For reference, the circuits in which no vertex repeats except the initial and terminal a are precisely

$$(a,a),\ (a,b,a),\ (a,c,b,a),\ (a,e,b,a),\ (a,e,d,a),$$
$$(a,b,e,d,a),\ (a,c,b,e,d,a),\ (a,e,c,b,a),$$
$$(a,e,d,b,a),\ (a,e,d,c,b,a).$$

The procedure also produces longer circuits such as (a,b,a,e,d,a); listing only the ten cycles above would not enumerate every circuit.

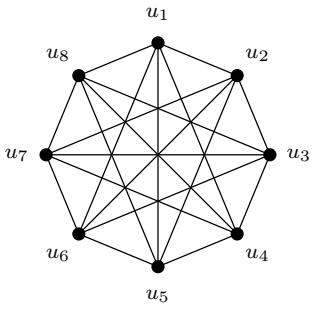
□

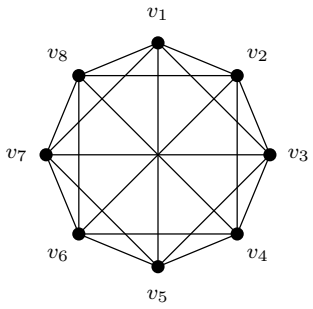
Problem 3. Show that isomorphism of simple graphs is an equivalence relation.

Solution. Let G, H, K be simple graphs. The identity map on $V(G)$ is an isomorphism from G to itself, so graph isomorphism is reflexive. If $f\colon V(G) \to V(H)$ is an isomorphism, then its inverse $f^{-1}\colon V(H) \to V(G)$ preserves both adjacency and nonadjacency; hence isomorphism is symmetric. If $f\colon V(G) \to V(H)$ and $g\colon V(H) \to V(K)$ are isomorphisms, then $g \circ f$ is a bijection that preserves adjacency in both directions. Thus isomorphism is transitive. Therefore it is an equivalence relation.

□

Problem 4. Determine whether the given pair of graphs is isomorphic. Exhibit an isomorphism or provide a rigorous argument that none exists.





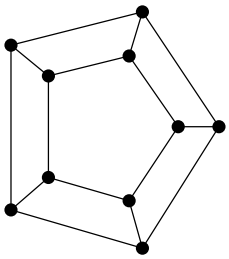
Solution. The graphs are *not* isomorphic. Read the vertex subscripts modulo 8. In the left graph, u_i is adjacent to $u_{i\pm1}$, $u_{i\pm3}$, and u_{i+4}. Hence its complement consists of

$$u_1u_3u_5u_7u_1 \qquad \text{and} \qquad u_2u_4u_6u_8u_2,$$

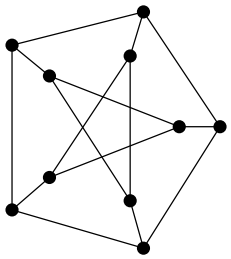
two disjoint 4-cycles. In the right graph, v_i is adjacent to $v_{i\pm1}$, $v_{i\pm2}$, and v_{i+4}, so its complement contains precisely the edges $v_iv_{i\pm3}$. Since $\gcd(3,8) = 1$, these form a single 8-cycle. The complements have different numbers of connected components and therefore cannot be isomorphic. It follows that the original graphs are not isomorphic.

□

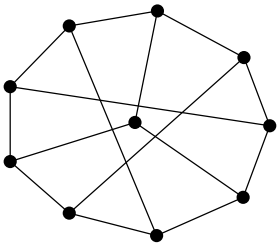
Problem 5. Determine which pairs of the following graphs are isomorphic. For each pair, either exhibit an isomorphism or provide a rigorous argument that none exists.



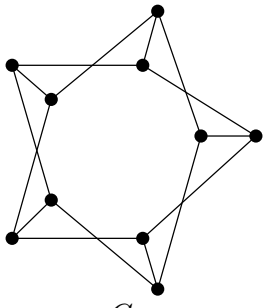
G_1



G_2



G_3



G_4

Solution. Only $G_2 \cong G_3$.

To exhibit an isomorphism, label the outer vertices of G_2 by x_0, x_1, x_2, x_3, x_4, clockwise from the top, and its corresponding inner vertices by y_0, y_1, y_2, y_3, y_4. In G_3, label the nine outer vertices by $z_0, \ldots, z_8$, clockwise from the top, and the central vertex by z_9. The following bijection preserves all 15 edges:

Vertex	x_0	x_1	x_2	x_3	x_4
Image	z_0	z_1	z_2	z_3	z_9
Vertex	y_0	y_1	y_2	y_3	y_4
Image	z_8	z_5	z_7	z_4	z_6

Indeed, the outer 5-cycle of G_2 maps to $z_0z_1z_2z_3z_9z_0$; the five spokes and the five inner edges also map to edges of G_3. Thus $G_2 \cong G_3$.

For the remaining pairs, G_1 contains a 4-cycle, while G_2 has none. Indeed, the two 5-cycles of G_2 are joined by a matching; a 4-cycle using two matching edges would require the same pair of indices to be adjacent in both 5-cycles, which is impossible (outer edges join consecutive indices, and inner edges join indices differing by two). Since $G_2 \cong G_3$, neither G_2 nor G_3 is isomorphic to G_1. Moreover, G_4 is bipartite: it consists of a 10-cycle together with five chords joining vertices at distance 5 along that cycle, all of which join opposite parts of the alternating bipartition. By contrast, G_1, G_2, and G_3 each contain a 5-cycle, so none is bipartite. Consequently, G_4 is not isomorphic to any of them.

□